

HEATER, VENTILATION & AIR CONDITIONING (HVAC)

07

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07-00 OUTLINE

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HVAC ABBREVIATION

dcf070000000t01

A/C	Air Conditioning
B+	Battery Positive Voltage
DEF	Defroster
HI	High
LOW	Low
M	Motor
MAX	Maximum
OFF	Switch Off
ON	Switch On
REC	Recirculate

HVAC FEATURES

dcf070000000t02

Improved air conditioning performance	Sub-cooling system to multi-flow condenser adopted
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HVAC SPECIFICATIONS

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Item		Specification
Cooling capacity (kW {kcal/h})		4.012 {3,450}
Airflow volume (during air conditioner operation)	Blower motor (m ³ /h)	450
	Electricity consumption (during air conditioner operation)	220
Blower motor	Magnetic clutch (W)	48
	Type	Sirocco fan
Refrigerant	Type	R-134a
	Regular amount (approx. quantity) (g {oz})	475 {16.8}
A/C compressor	Type	Swash-plate
	Discharge capacity (ml {cc, fl oz})	154 {154, 5.16}
	Max. allowable speed (rpm)	7,000
	Lube oil Type	FD46XG
	Sealed volume (approx. quantity) (ml {cc, fl oz})	180 {180, 6.08}
Magnetic clutch clearance mm {in}		0.35—0.75 {0.014—0.029}

07

07-00-1

OUTLINE

Item		Specification
Condenser	Type	Multiflow (sub-cooling type)
	Radiated heat (kW {kcal/h})	10.46 {8,996}
	Receiver/drier capacity (ml {cc, fl oz})	220 {220, 7.44}
	Desiccant	Synthetic zeolite
Expansion valve	Type	Internal pressure equalizar
Evaporator	Type	Single-tank drawn cup
Refrigerant pressure switch	Type	Dual-pressure type
	Operating pressure	0.176—0.216 {1.79—2.20, 25.46—31.28} 2.94—3.34 {30.0—34.0, 427—483} ON 0.02 {0.20, 2.84} or less 0.39—0.79 {3.98—8.05, 57—114} OFF
	(MPa {kgf/cm ² , psi})	
Fusible plug	Melting point (°C {°F})	102—107 {216—224}

BASIC SYSTEM

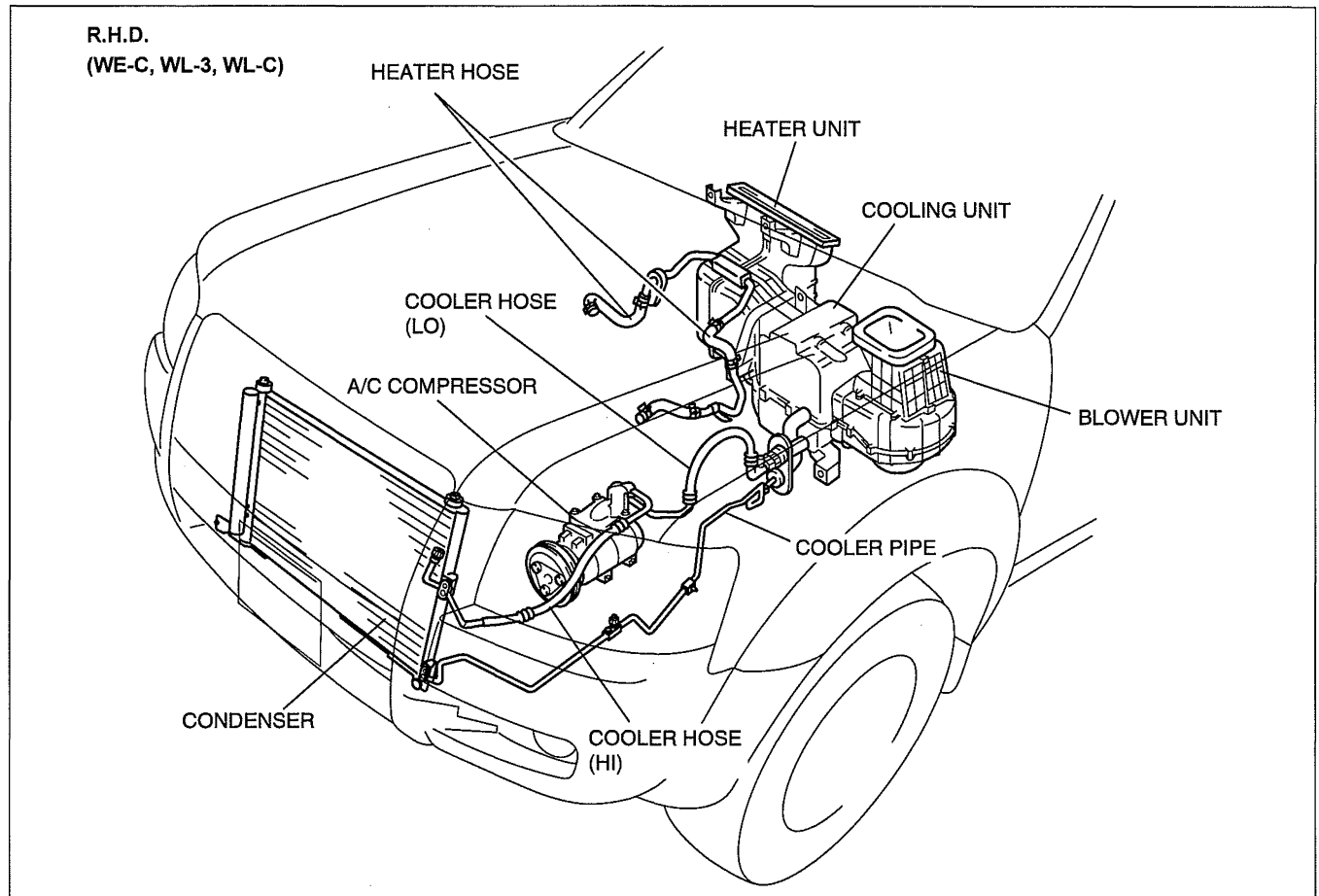
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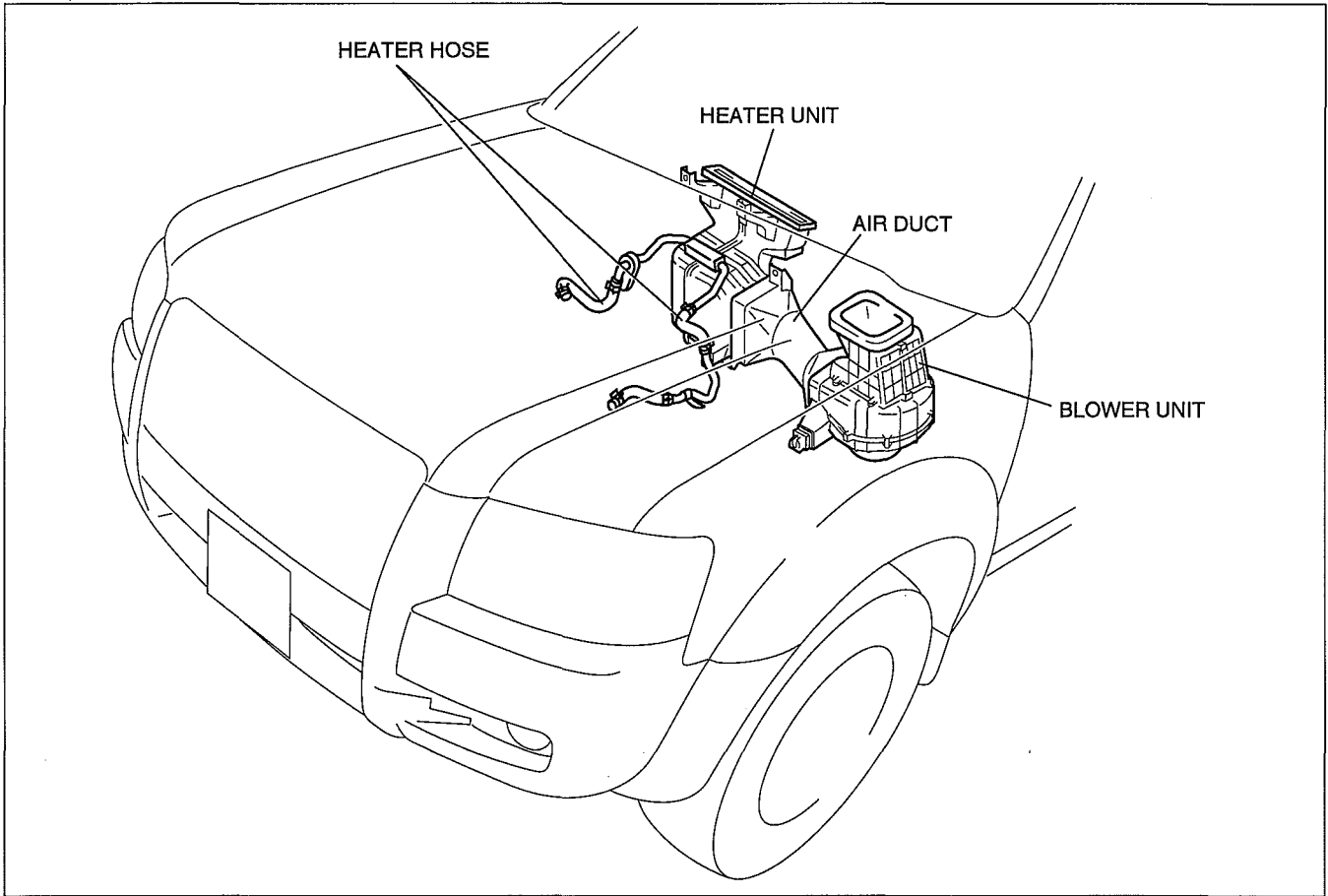
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BASIC SYSTEM

Without A/C



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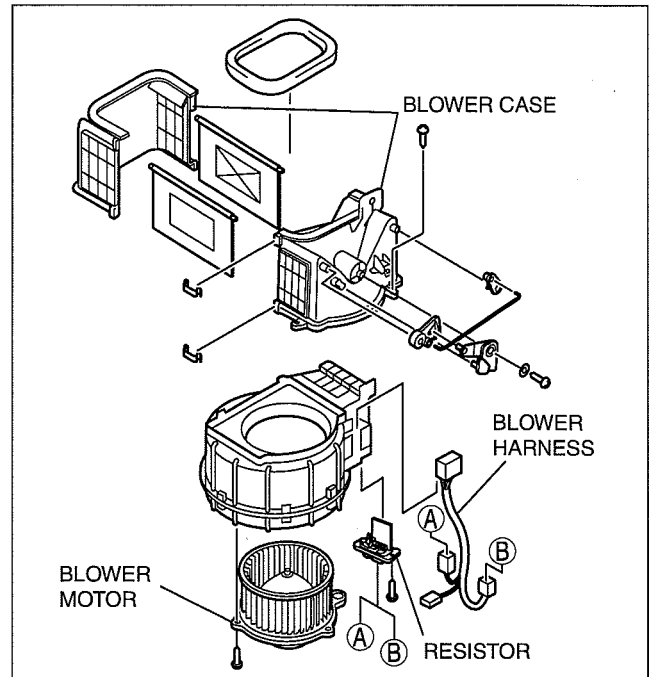
BASIC SYSTEM

BLOWER UNIT CONSTRUCTION/OPERATION

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- The blower unit consists of a blower motor, resistor, blower harness, blower case.

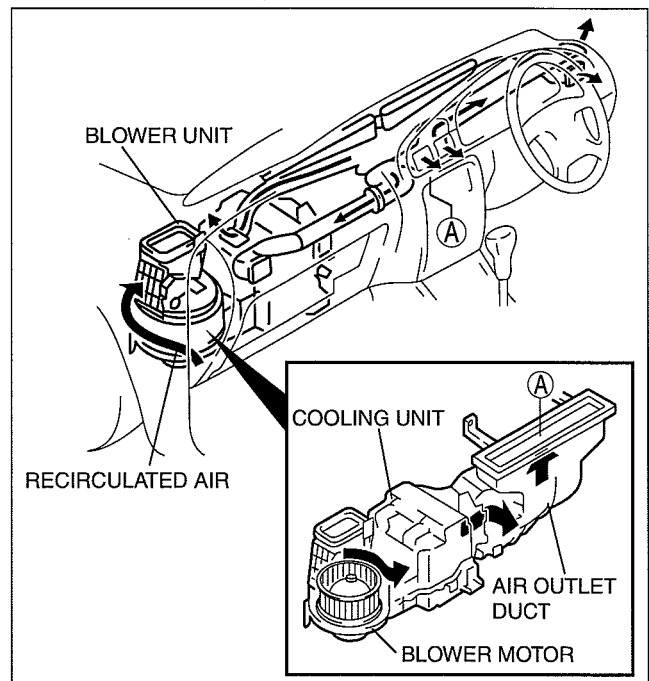
Construction



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Operation

- The blower unit blows recirculated air into the passenger compartment.



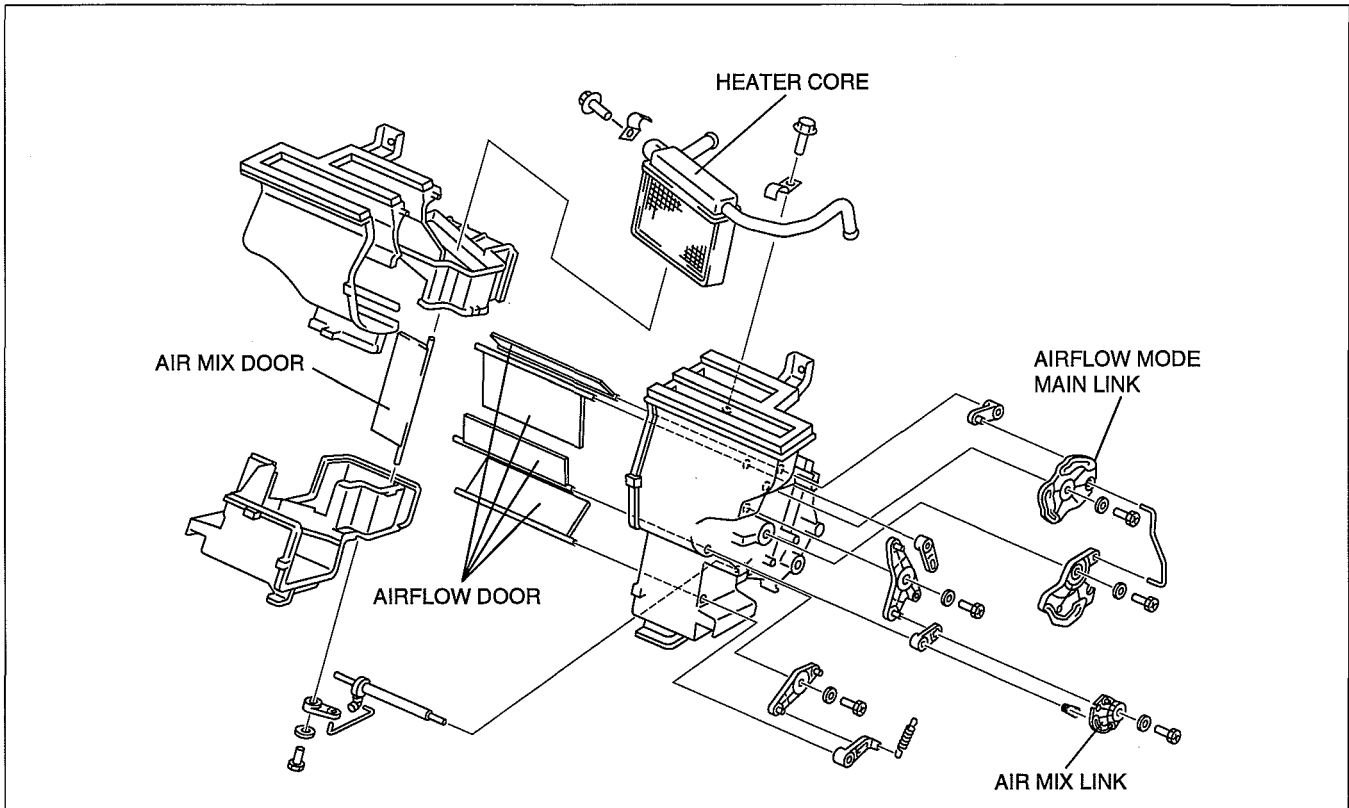
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BASIC SYSTEM

HEATER UNIT CONSTRUCTION

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- The heater unit consists of a heater core, airflow mode main link, air mix link etc.
- The heater unit regulates the airflow mode and the outlet airflow temperature.



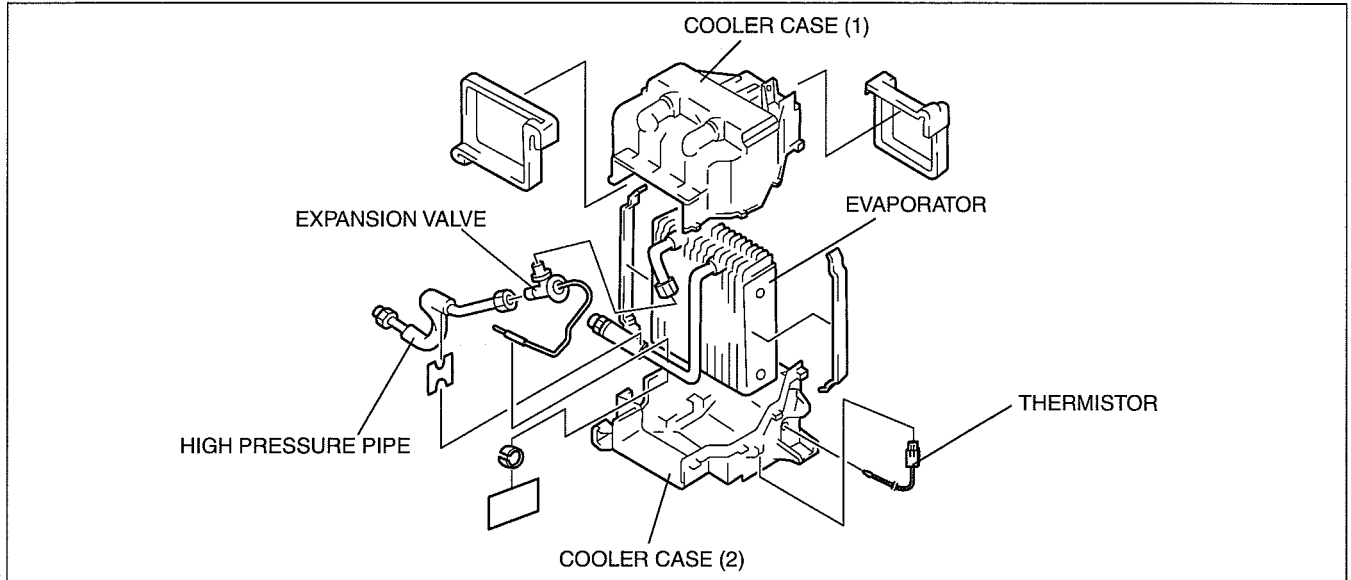
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BASIC SYSTEM

COOLING UNIT CONSTRUCTION/OPERATION

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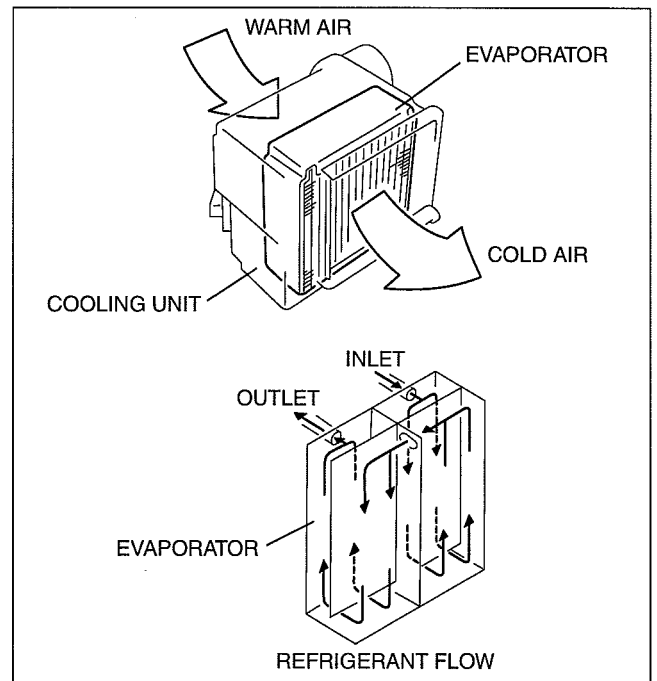
- The cooling unit consists of an evaporator, expansion valve, high pressure pipe, cooler case.
- The cooling unit cools and dehumidifies the airflow from the blower unit.
- The thermistor is built into the cooling unit and controls the A/C compressor.



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Evaporator

- The evaporator vaporizes the refrigerant flowing inside it. The latent heat of the vaporized refrigerant cools the air passing through the evaporator. At the same time, moisture in the air condensates. In this way, dehumidified air is supplied to the passenger compartment.
- The evaporator is a single tank drawn cup type. Because refrigerant flows through several separate tubes, flow path resistance is kept low and the surface area for heat exchange is increased, improving air conditioning efficiency.

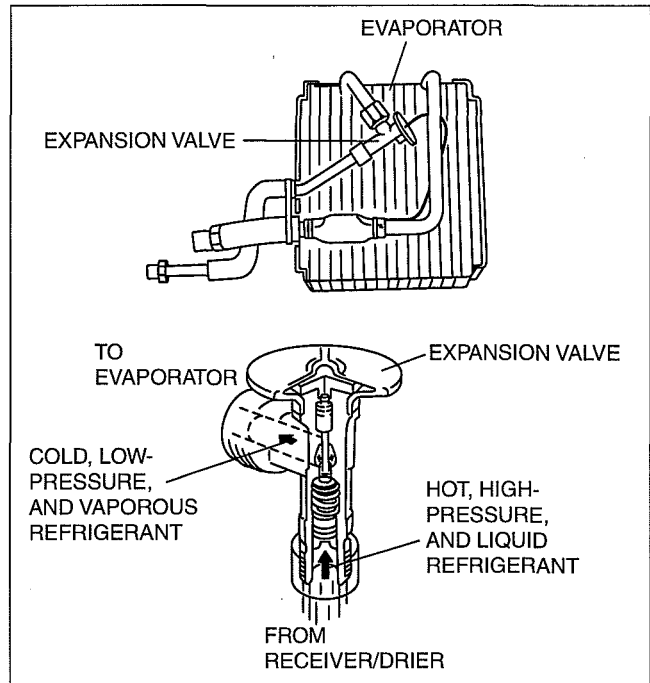


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BASIC SYSTEM

Expansion valve

- The expansion valve causes a sudden decrease in the pressure of the liquid refrigerant. This atomizes the refrigerant, making it easier for the evaporator to vaporize it. The expansion valve also regulates the flow volume of the refrigerant sent to the evaporator.
- An internal pressure equalizer expansion valve is used. It regulates the flow volume of the refrigerant supplied to the evaporator so that the heat absorption process of the evaporator is utilized efficiently.

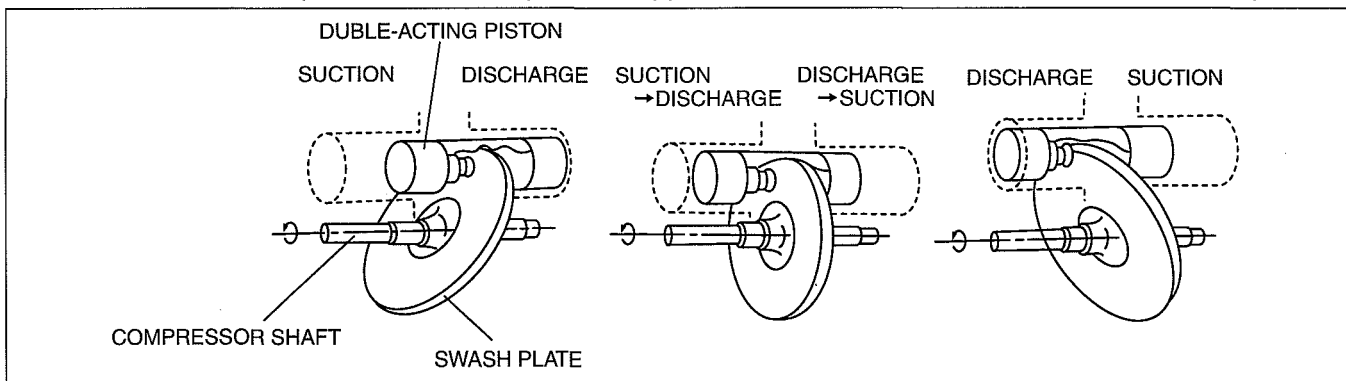


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A/C COMPRESSOR CONSTRUCTION

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- The A/C compressor compresses the low-pressure gaseous refrigerant vaporized from the evaporator into a high-pressure gas so that the condenser can liquidize it more easily. A swash-plate of the A/C compressor is used to reduce weight and vibration and to improve driving performance when the A/C compressor is operating.
- The A/C compressor is belt-driven via the engine crankshaft pulley.
- Five double-acting pistons positioned axially around the compressor shaft are actuated by a swash plate that is pressed onto the compressor shaft. The swash plate changes the rotating action of the shaft to reciprocating movement of the five pistons. This driving force is applied to the midpoint of each of the double-ended pistons.

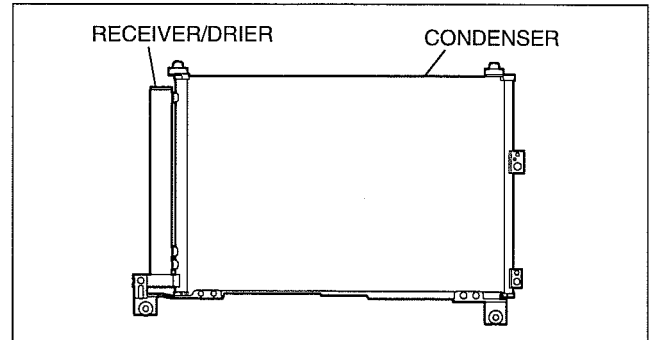


DBG711ZTB008

BASIC SYSTEM

CONDENSER CONSTRUCTION

- The receiver/drier, main condenser, and sub-condenser are integrated into the sub-cooling system multi-flow condenser.
- The sub-cooling system multi-flow condenser sends refrigerant cooled in the main condenser to the receiver/drier and re-cools it in the sub-condenser for a more efficient cooling process.



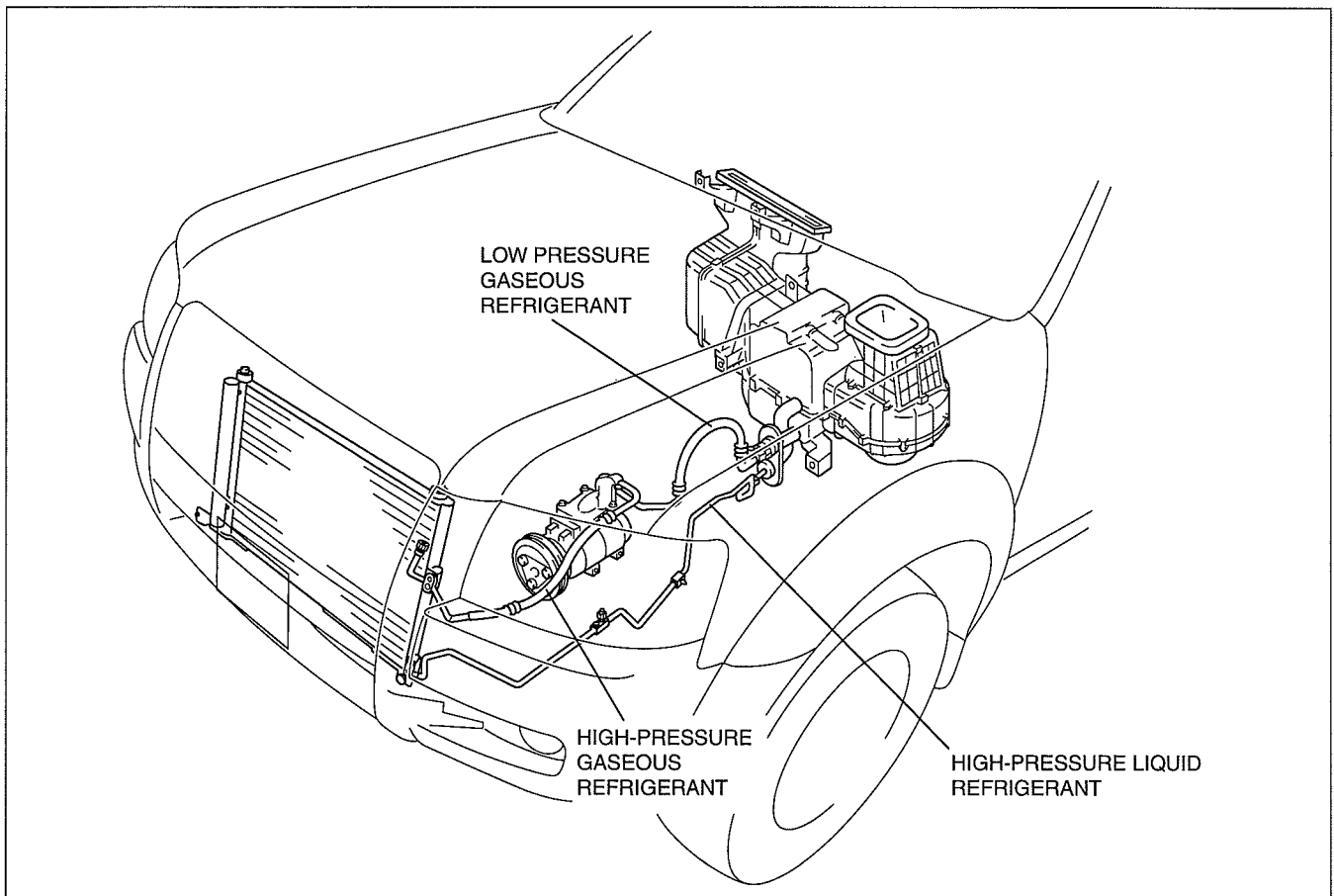
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REFRIGERANT LINE CONSTRUCTION

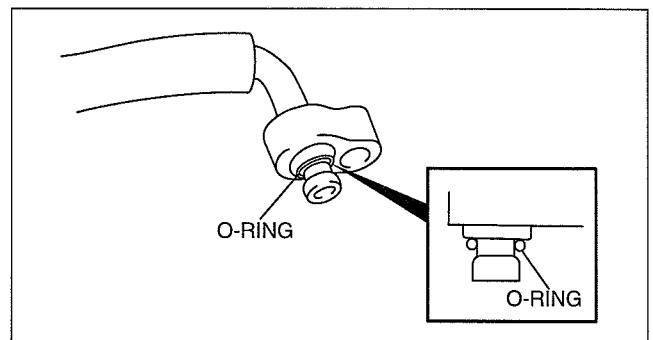
- Cooler pipes and cooler hoses are refrigerant passages.
- Refrigerant flows inside of each pipe and hose in the form of high-pressure gas from the A/C compressor to condenser, in the form of high-pressure liquid from the condenser to the expansion valve, and in the form of low-pressure gaseous from the expansion valve to the A/C compressor.
- The diameter of each pipe is different according to the state of the refrigerant.



DBG7112TB022

Piping Connection

- To prevent O-rings from falling off while piping work is being performed, stay-on O-rings are used at the piping connections.



DBG7112TB015

(11)

(12)

(13)

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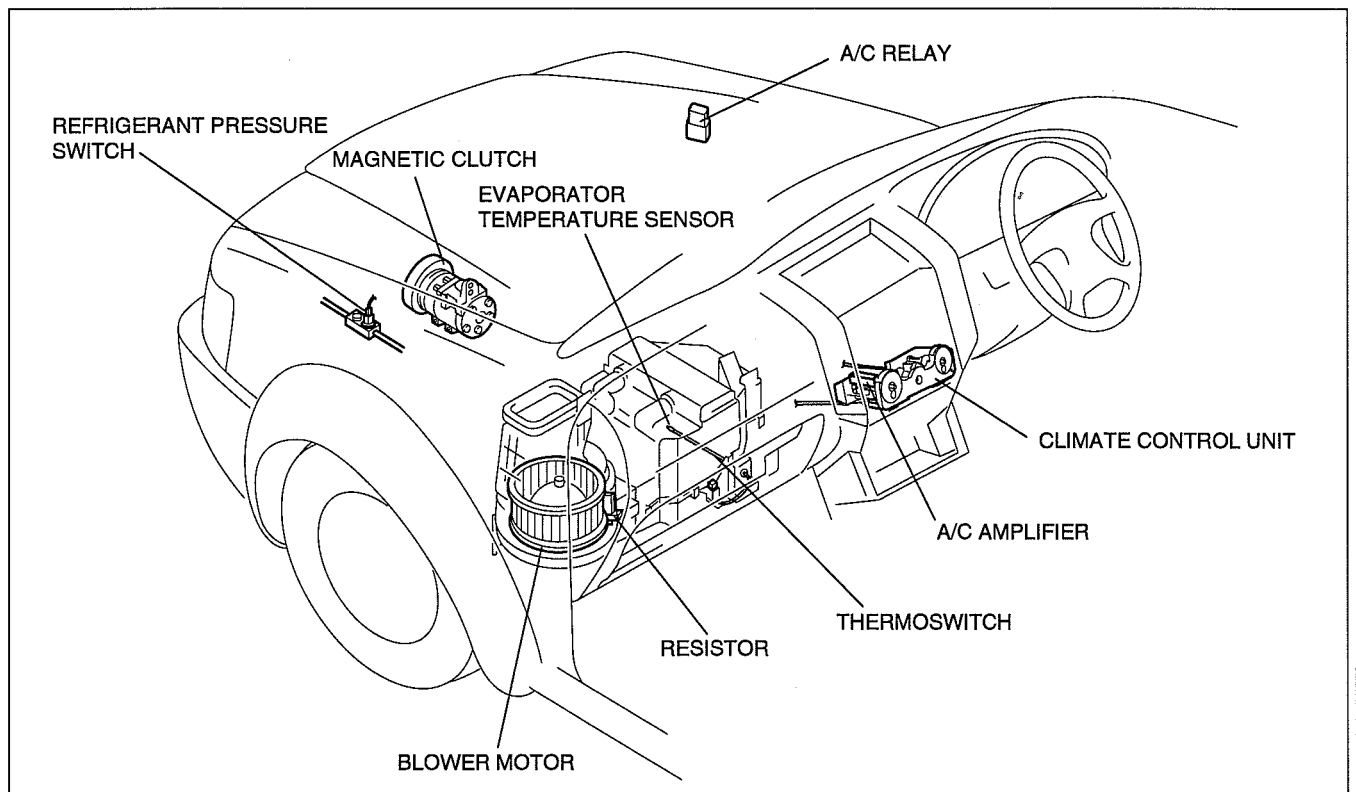
CONTROL SYSTEM OUTLINE

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Improved operability	Climate control unit with enlarged operation dial and switch adopted
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CONTROL SYSTEM LOCATION INDEX

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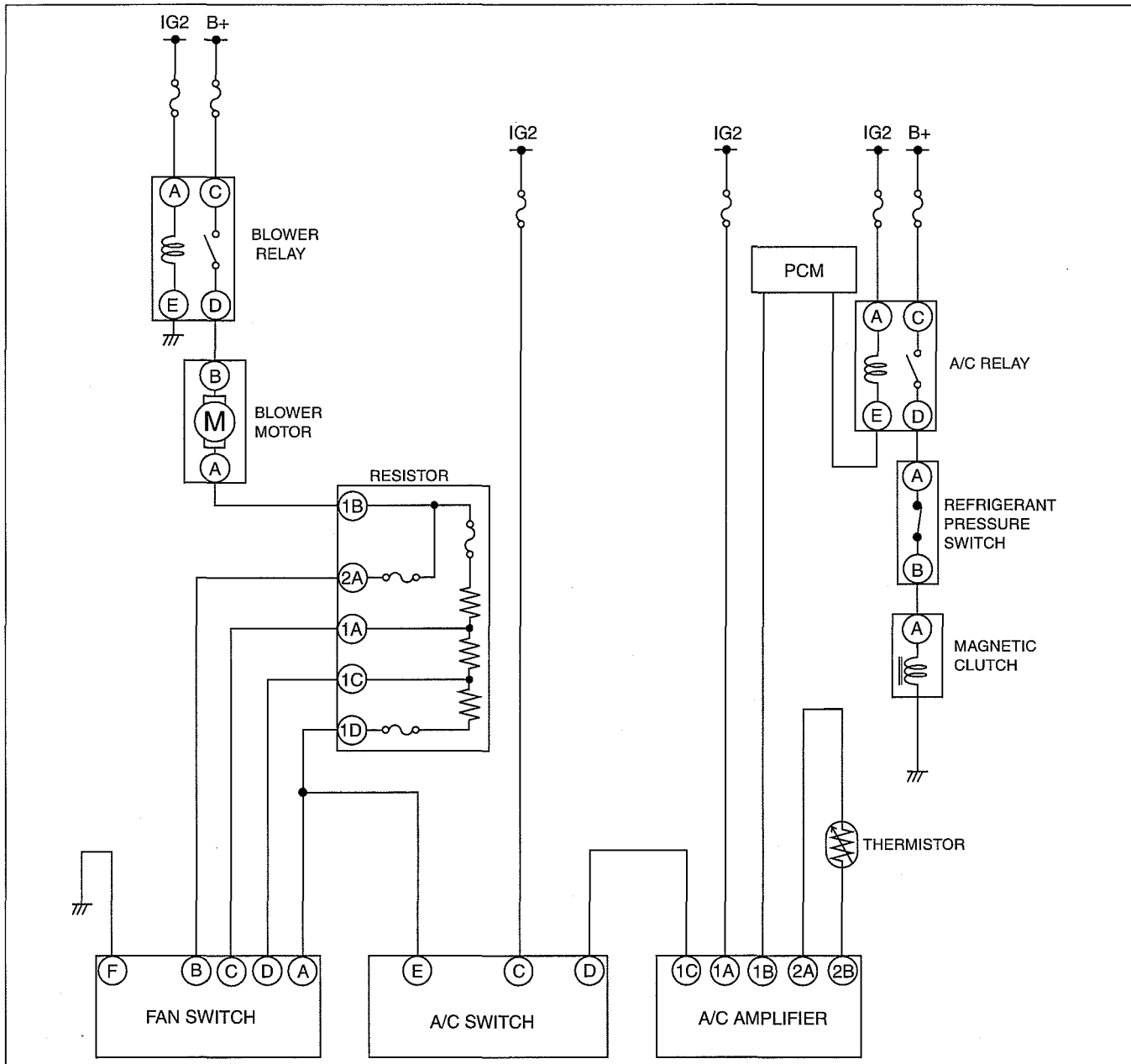


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CONTROL SYSTEM

CONTROL SYSTEM WIRING DIAGRAM

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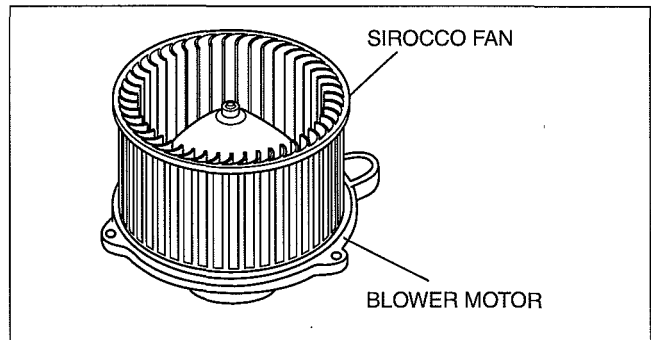


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BLOWER MOTOR CONSTRUCTION

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- The blower motor rotates the blower fan to blow air into the passenger compartment.
- The blower fan is a sirocco-type fan.
- To prevent a blower motor from burning by overheating, the blower air always blows inside of the blower motor from the air duct when the blower motor is operated.



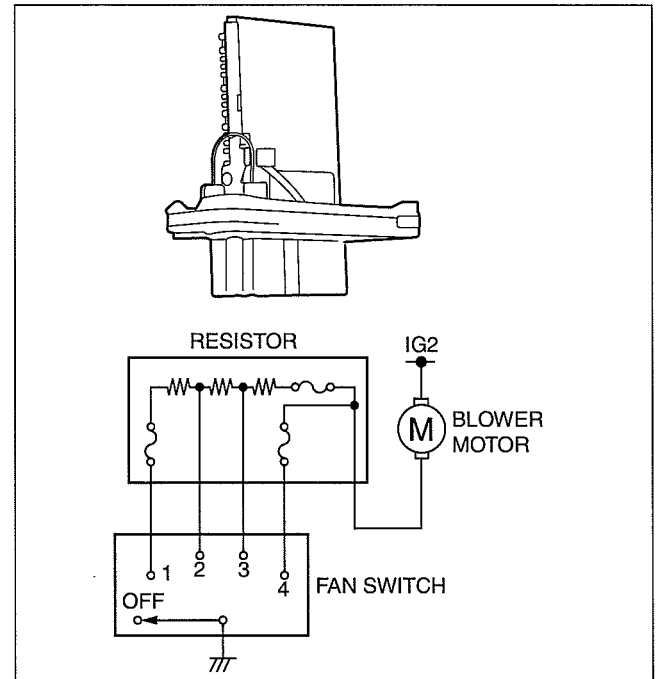
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CONTROL SYSTEM

RESISTOR CONSTRUCTION

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- The resistor controls the voltage applied to blower motor, corresponding to fan switch operation. The resistor controls the rotation speed (airflow volume) in 4 stage.

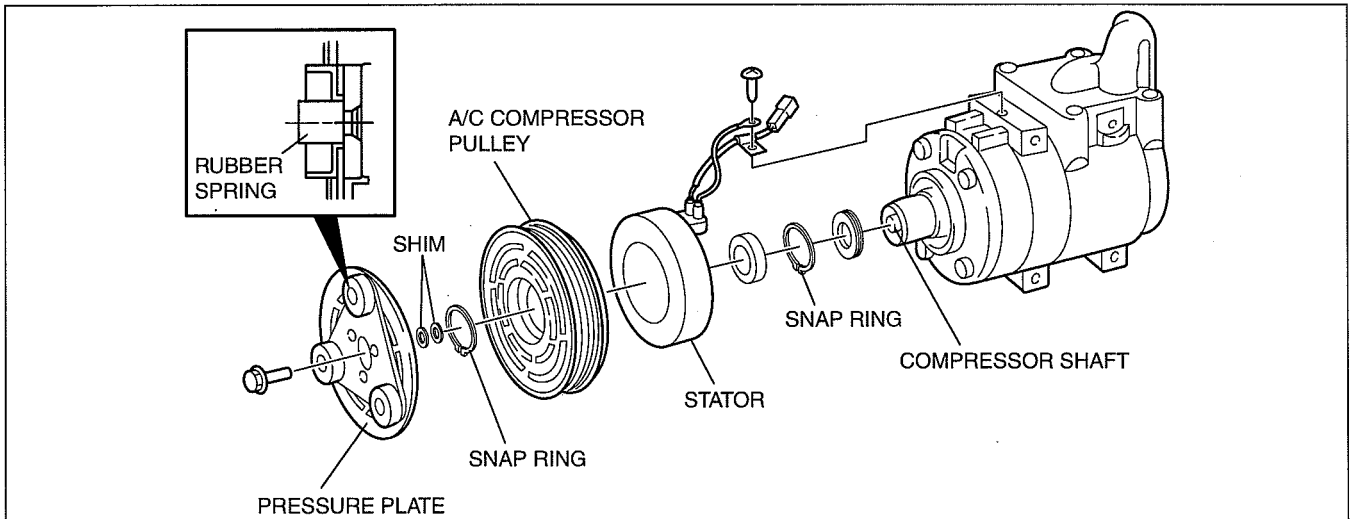


DBG740ZTB010

MAGNETIC CLUTCH CONSTRUCTION

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- A magnetic clutch drives the compressor shaft.
- When voltage is applied to the stator, the magnetic force locks the pressure plate and the pulley together as one unit. As a result the compressor shaft then turns with the pulley.
- When voltage is removed from the stator, the pressure plate returns from the pulley by the rubber spring force. As a result the the compressor shaft does not turn.



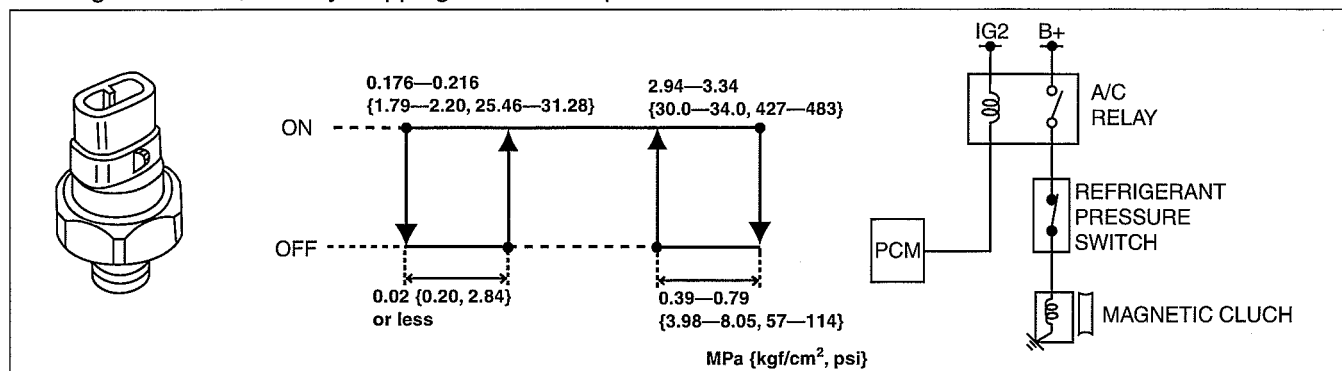
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CONTROL SYSTEM

REFRIGERANT PRESSURE SWITCH CONSTRUCTION

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- The refrigerant pressure switch is fitted to the cooler pipe and senses the refrigerant pressure.
- A dual-pressure refrigerant pressure switch is used to respond to both abnormally high and abnormally low pressures.
- When pressure is abnormal, the refrigerant pressure switch turns off and cuts the voltage sent from the magnetic clutch, thereby stopping the A/C compressor.

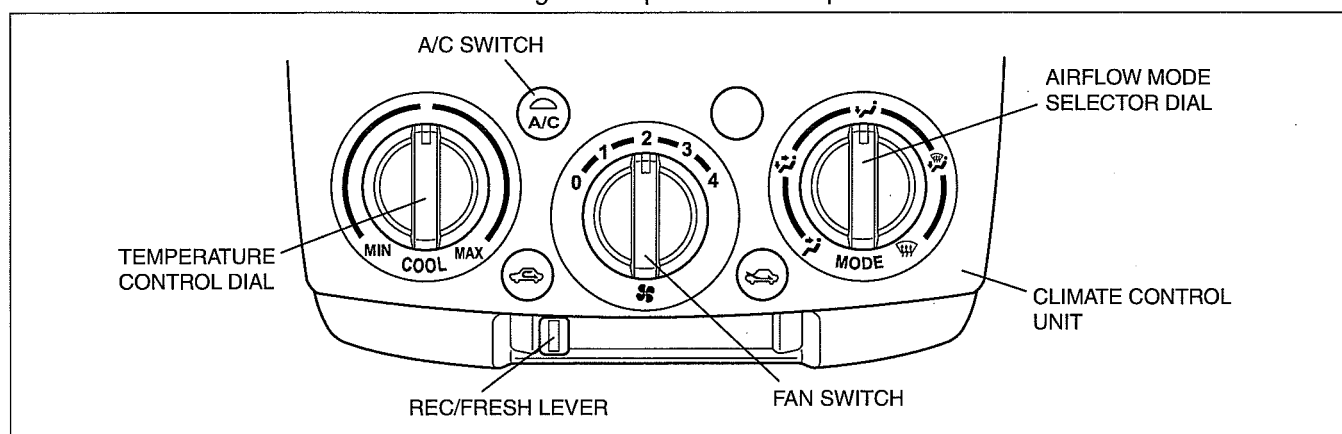


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CLIMATE CONTROL UNIT CONSTRUCTION

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- Each switches and dials have been enlarged to improve ease of operation.



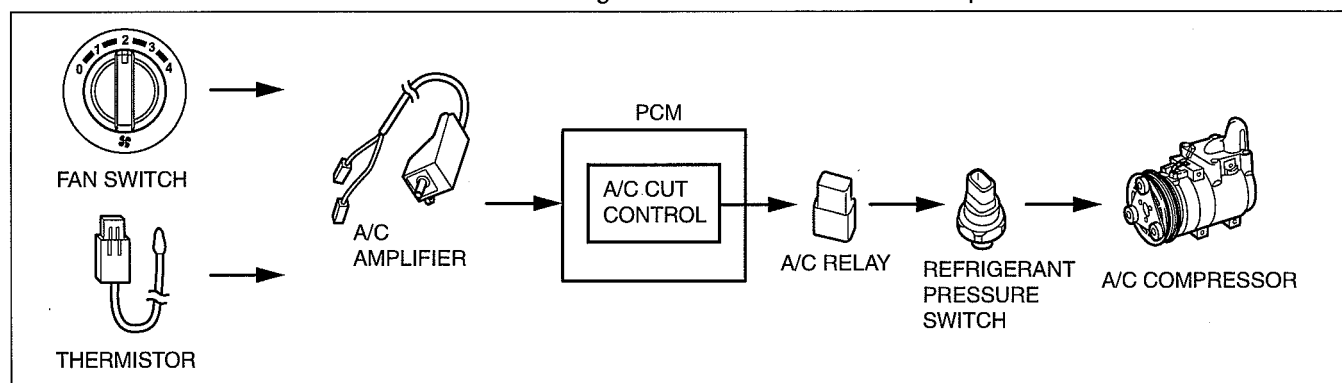
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MANUAL AIR CONDITIONER CONTROL SYSTEM

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Block Diagram

- The fan switch and thermistor sends an A/C signal to the PCM via the A/C amplifier.

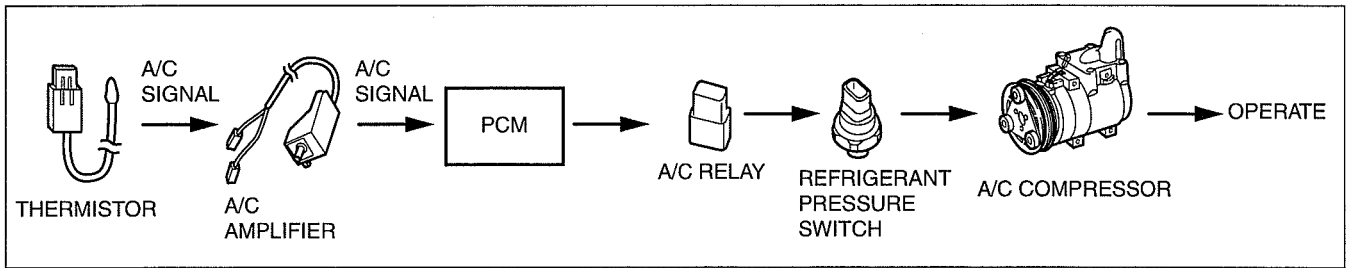


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CONTROL SYSTEM

A/C Compressor Control

- The A/C signal from the thermistor and A/C amplifier controls the magnetic clutch via the PCM, A/C relay and refrigerant pressure switch.



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